HK IMO Training 2019-2020
Problem Set 30
For 6 June 2020

Problem 117. Euclid placed 8 points on the plane. Then he writes down all the areas of the triangles (a total of 56 numbers) constructed by these points.
Prove that he can always assign positive or negative signs to these numbers such that the sum of these numbers becomes zero.

Problem 118. In $\triangle ABC$, with $AB < AC$, let

- D and E be points on the sides AB and AC, respectively, with $D, E \neq A, B, C$, such that $CA = CD$ and $BA = BE$;
- Ω be the circumcircle of $\triangle ADE$;
- P be reflection of A about BC;
- $X = PD \cap \Omega$ and $Y = PE \cap \Omega$ with $X \neq D$ and $Y \neq E$.

Prove that the two lines BX and CY intersect at a point lying on Ω.

Problem 119. Let $S = \{1, 2, \ldots, 999\}$. Suppose a function $f\colon S \to S$ satisfies the following condition:

$$f^{n+f(n)+1}(n) = f^{nf(n)}(n) = n \quad \text{for every } n \in S.$$

Prove that there exists an element $a \in S$ which satisfies the condition $f(a) = a$.
Here $f^k(n)$ means the iteration of the function $f(n)$ k-times, i.e., $f^2(n) = f(f(n))$, $f^3(n) = f(f(f(n)))$, etc.

Problem 120. The sequence $a_1, a_2, \ldots$ satisfies the equality

$$\sum_{k=1}^{n} a_{\left\lfloor \frac{n}{k} \right\rfloor} = n^{10} \quad \text{for all positive integers } n.$$

Given an arbitrary positive integer c, prove that

$$\frac{c^{a_n} - c^{a_{n-1}}}{n}$$

is an integer for each $n > 1$.
Here $\lfloor x \rfloor$ denotes the greatest integer not exceeding x.